

Sanjeev Krishnan Kamalamurugan

📍 Portland, OR ✉ sanjeev24433@gmail.com ☎ +1 (503)-504-4109 [in Sanjeev Krishnan](#) [Github Portfolio](#)

OBJECTIVE: Seeking New College Graduate roles in ASIC Design and Verification Engineering.

EDUCATION

M.S., Electrical and Computer Engineering GPA: 3.6 / 4.0

Portland State University, Portland, Oregon (Graduated: June 2026)

Coursework: Microprocessor System Design, Computer Architecture, Advanced Computer Architecture, System Verilog, Pre-Silicon Validation, Assertions Based Verification, Formal Verification, Post-Silicon Validation

B.E., Electronics and Communication Engineering GPA: 8.3 / 10

Anna University, Chennai, India (Graduated: April 2023)

TECHNICAL SKILLS

- **Hardware Description Languages:** Verilog, SystemVerilog, System Verilog Assertions (SVA)
- **Programming & Scripting Languages:** C, C++, OOPs, Python (Basics)
- **Computer Architecture:** RISC-V ISA, DDR SDRAM, Cache Coherence, Virtual Memory, Branch Prediction, Pipelining, Out-of-Order Execution
- **Verification Methods:** UVM, Constrained Random Tests, Functional Coverage, Assertion Based Verification
- **Protocols & Interfaces:** APB, AXI, MESIF, UART, SPI
- **Tools:** QuestaSim, Synopsys VC Formal, Cadence JasperGold, Vivado, Vitis, Git, Visual Studio Code, Linux

EXPERIENCE

ECE 571 - System Verilog Course Grader Jan 2026 – Mar 2026

Portland State University, Portland

- Evaluating **RTL Design and verification assignments and projects** focused on System Verilog testbenches.
- Providing **technical feedback on digital design, verification methodology, and simulation** concepts.

PROJECTS

Floating Point Unit | [System Verilog, RTL Design, UVM]

- Developed **UVM-based** constrained random verification environment including **scoreboards, monitors, and coverage models** for **IEEE 754-compliant** Floating Point Unit for Arithmetic Operation.
- Generated **300+ stimulus** through constraints and randomization verifying corner cases like **Infinity, NaN, and subnormals** to ensure full compliance of the unit, achieving a function coverage of **100%**.

Design and Verification of Asynchronous FIFO | [System Verilog, RTL Design, Assertions]

- Verified a **45-deep asynchronous FIFO** with **gray-code pointers and synchronizers** using SystemVerilog Assertions, achieving **100% functional coverage** through well-defined cover properties.
- Validated **data integrity, metastability handling, Clcok Domain Crossing(CDC) and boundary conditions** by checking **overflow, underflow, and timing hazard** scenarios across multiple clock domains.

2 x 2 Router Design and Verification | [System Verilog, RTL Design, Assertions]

- Design and Verification of **packet-based 2x2 router RTL** using dynamic simulation and Assertions. Developed testbench architecture and analyzed waveforms for end-to-end packet transfer and control logic.
- Identified and **fixed RTL bugs using counterexample traces**, wrote **parameterized assertions, assumptions, and cover properties** to prove protocol correctness using VC Formal and JasperGold.

Design and Verification of Cache Controller | [System Verilog]

- Designed and verified **L2 cache** featuring **16-way associativity** and **64-byte line size** using **write-back and write-allocate** policies. Implemented **P-LRU policy** and **MESI protocol** for eviction and coherence.
- Verified with comprehensive test cases achieving performance of **99.6% hit ratio**.

RISC-V RV32IM Simulator | [C, RISC-V Assembly]

- Implemented **5-stage pipelined RISC-V ISA** with Instruction Fetch, Decode, Execute, Memory and Writeback in single cycle model, managing the **64KB memory constraint, program counter and stack pointer**.
- Verified through RISC-V Assembly language by generating **memory image files** using **RISC-V cross compiler**.

CERTIFICATION

Accelerated Computing in Modern CUDA C++

Issued by Nvidia (July 2025)

- Completed hands-on training in CUDA C++ for GPU-accelerated computing, covering parallel programming, memory optimization, kernel execution, thread hierarchy, and performance profiling.